CHAPTER 1: TURBOPUMP & HYDRAULIC ACTUATION OVERVIEW

This technical manual specifies mandatory overhaul tolerances, MIL-SPEC fastener torque schedules, Stage-2 high-pressure relief valve schematic callouts, and Stage-3 cryogenic labyrinth seal clearances.

SUMMARY OF CRITICAL SECTIONS IN THIS VOLUME:

- Page 2 (Archive P. 082): Section 4.2 Primary Hydraulic Manifold Torque Table (MS90725-114)
- Page 3 (Archive P. 014): Figure 4-9 Stage-2 Relief Valve CAD Schematic (O-Ring AN6227-23)
- Page 4 (Archive P. 1184): Table 19-14 Cryogenic Labyrinth Seal Clearances & Footnote 17

SECTION 4.2: HYDRAULIC MANIFOLD TORQUE & FASTENER SPECS

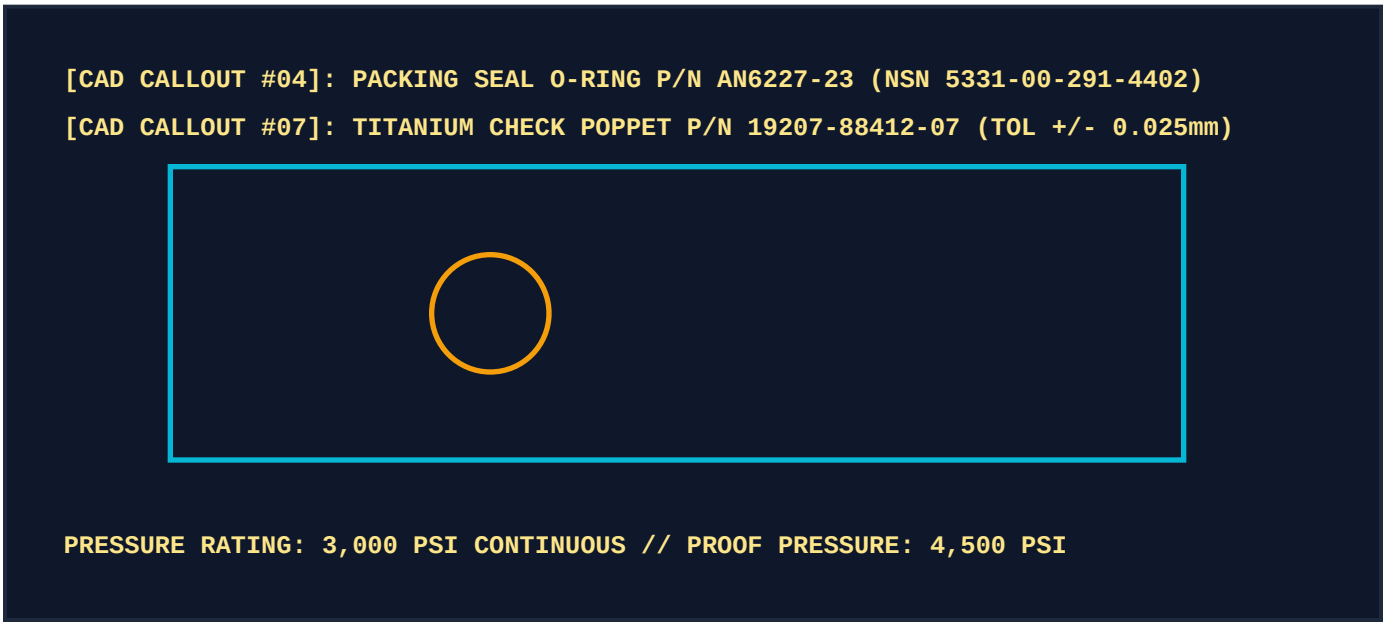
Per Section 4.2 Torque & Fastener Specifications, the primary hydraulic manifold M12 flange bolts require MIL-SPEC fastener Part No. MS90725-114 torqued to 45.0 Nm +/- 1.5 Nm (33 lb-ft) using an 8-point cross-star sequence. Verify calibration of digital torque wrench prior to assembly.

TABLE 4-3: PRIMARY HYDRAULIC MANIFOLD FLANGE FASTENERS (ARCHIVE PAGE 82)

Assembly Location	MIL-SPEC P/N	Nominal Torque	Sequence
Primary Manifold M12	MS90725-114	45.0 Nm +/- 1.5 Nm	8-Bolt Star
Bypass Housing M8	AN4-12A	19.5 Nm +/- 0.8 Nm	4-Bolt Cross
Return Flange M10	NAS6204-18	31.0 Nm +/- 1.0 Nm	6-Bolt Star

FIGURE 4-9: STAGE-2 HIGH-PRESSURE RELIEF VALVE CAD SECTION

On the Stage-2 High-Pressure Relief Valve CAD assembly drawing (DWG-88412-CAD Rev C, Page 14), Callout #04 specifies fluorocarbon packing O-Ring AN6227-23 (NSN 5331-00-291-4402) rated for 3,000 PSI continuous operating pressure. Callout #07 specifies Titanium Check Poppet P/N 19207-88412-07.



CHAPTER 19.8: CRYOGENIC TURBOPUMP LABYRINTH SEAL CLEARANCES (PAGE

Under -196 C cryogenic thermal shock testing per MIL-STD-810H Method 503.7, Stage-3 turbopump labyrinth seals must strictly maintain the axial play clearances specified in Table 19-14 below.

TABLE 19-14: CRYOGENIC TURBOPUMP SEAL CLEARANCES (ARCHIVE PAGE 1184)

Seal Stage	Thermal Regime	Max Axial Play	Backing Shim P/N
Stage-1 Ambient	+20 C to +140 C	0.045 mm (+/-0.005)	MS90725-109
Stage-2 Interstage	-65 C Steady State	0.030 mm (+/-0.004)	MS90725-112
Stage-3 Labyrinth	**-196 C Cryo Shock**	**0.018 mm (+/-0.002)***	**NAS1149-F0832P [17]**

FOOTNOTE 17 (MANDATORY CRYOGENIC SUPERSESSION - PAGE 1184):

Under -196 C cryogenic thermal shock, Stage-3 labyrinth seal maximum allowable axial play is strictly 0.018 mm (+/- 0.002 mm). Standard carbon-steel washer MS90725 is SUPERSEDED by cryo-stabilized high-nickel Inconel shim washer P/N NAS1149-F0832P per DWG-99401 Rev H.